

Regime switches and commonalities of the cryptocurrencies asset-class

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Abstract

In this paper we test for regime changes in the price dynamics of Bitcoin, Ethereum, Litecoin and Monero, as representatives of the cryptocurrencies asset class. Data are observed daily from January, 1, 2016 to October, 15, 2019. Best specifications within Gaussian and Autoregressive Hidden Markov models for price differences are selected through the AIC and BIC information criteria by considering up to four hidden regimes. The empirical results suggest that at most three common states may be considered for the basket of cryptocurrencies under investigation; a fourth state may be relevant as an added factor to the dynamics description of the individual cryptocurrencies rather than to the whole basket. Finally, we test the out-of-sample performance of estimated regime switching models; optimal results, in terms of RMSE and correlation between predicted and real values, are obtained in the case of two common or three individual regimes.

Keywords Cryptocurrencies · Regime Switching · Hidden Markov Models · Forecasting analysis

JEL Classification C22 · C52 · C53 · C58

1 Introduction and literature review

Cryptocurrencies are used both as media of exchange and as components of investment portfolios. In both cases, it is crucial to analyse the econometric

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behavior of prices and returns of cryptocurrencies in order to make forecasts. The econometrics behaviour of cryptocurrencies, in particular Bitcoin, has been the subject of many papers in the last few years.

A relevant stream of research aims at understanding which external factors drive cryptocurrencies' price changes. In Dyhrberg (2016) Bitcoin returns are modeled depending on explanatory variables related to the financial markets, among which the price of gold, gold futures, the Financial Times Stock Exchange Index. The, somehow controversial, relationship with Gold is also recently investigated in Adebola et al. (2019), where the authors find a limited connection between cryptocurrencies and gold prices, when applying fractional integration and cointegration techniques: gold prices exhibit a mean reverting behavior while the majority of analyzed cryptocurrencies do not. It has also been argued that cryptocurrencies serve as an hedge against the risk of stock markets or as a safe-haven, see among others, Tiwari et al. (2019), where the authors investigate the time-varying correlations and conditional volatility between the cryptocurrency and S&P 500 index returns.

Kristoufek (2015) investigates the relationship between Bitcoin price and crypto-related factors, such as the hashrate, the mining difficulty, the number of transactions and the total number of Bitcoin in circulation. Ciaian et al. (2016); Figà-Talamanca and Patacca (2019); Ahn and Kim (2019); Eom et al. (2019) address the question whether Bitcoin returns and volatility are influenced by investor attention, sentiment or by specific measures of market attractiveness. Indeed, the cryptocurrency sector seems to be driven by periods of high enthusiasm followed by periods of skepticism; structural breaks in the dynamics as well as bubble effects have been detected, among others, in Garcia et al. (2014); Cheah and Fry (2015); Fry and Cheah (2016); Corbet et al. (2018); Bouri et al. (2019); Cretarola and Figà-Talamanca (2019); Chaim and Laurini (2019).

Most of the above contributions focus on Bitcoin which is usually considered as a benchmark for the whole sector. Indeed, the relationship between different cryptocurrencies is very strong and empirical facts for Bitcoin are also displayed by the others. For instance, in Yaya et al. (2019), the dependence between the price dynamics of several altcoins and that of Bitcoin is investigated, by applying the theory of fractional cointegration; interestingly, they evidence a reduction in the cointegration after the Bitcoin bubble burst at the end of 2017. A slightly out of the crowd results is given in Kumar and Anandarao (2019), by applying a Dynamic Conditional Correlation Integrated GARCH model and wavelet methods, where a weak volatility spillover is identified across the returns of Bitcoin, Ethereum, Litecoin and Ripple, as an evidence that the volatility of a specific cryptocurrency is affected mostly by its own fluctuations.

Motivated by the above research findings, we investigate whether Bitcoin and other cryptocurrencies share common regimes within the same time-period. This fact would further strengthen the interdependence between cryptocurrencies and may be exploited in an investment portfolio perspective. For the Bitcoin case, Garcia et al. (2014); Ardia et al. (2019); Koutmos (2019) prove the presence of multiple regimes; here, we extend the analysis in a multivariate setting by considering several cryptocurrencies.

In order to conduct our analysis, we select, among the 20 cryptocurrencies with highest market cap according to <https://coinmarketcap.com/>, those which existed and traded for at least three years: Bitcoin, Ethereum, Litecoin and Monero.

We estimate Gaussian and Autoregressive Hidden Markov models for price differences up to four states, characterized by varying mean gain and volatility, both in a univariate and multivariate setting and show that AIC and BIC information criteria strongly improve when allowing at least two regimes. In that case they may be safely interpreted as a low volatility and a high volatility state where high volatility is associated with negative mean differences, while low volatility is associated with positive mean differences. The probability of changing regime is lower than the persistence probability for each of the states and, notably, the high volatile-low mean regime has dominated the stable state in term of average sojourn time within the time period under analysis.

When cryptocurrencies are considered individually, the information criteria are optimized in case three or four states are allowed; in the multivariate approach, when regimes are constrained to correspond across the whole basket, optimal selection leads to a three states model specification. In this case the low-volatility state is trailed with two high-volatility periods, respectively associated to (high) positive mean gains and negative mean gains, which may be interpreted as optimistic and pessimistic high volatility states.

Finally, we analyze the out-of-sample performance of the estimated models on 100 moving windows/test dates and show that Hidden Markov Models (HMMs) do have forecasting power for the four cryptocurrencies, both in terms of Root Mean Squared Error (RMSE) and correlations between predicted and true values of price differences. It is worth noticing that best performance is obtained when considering three single or two common regimes; hence the results in this case are more conservative than in the AIC-BIC selection. This finding may be interpreted by arguing that the presence of three single or two common regimes is stable in time while additional regimes may be relevant only for specific sub-periods.

The rest of the paper is organized as follows: Section 2 is devoted to a preliminary statistical analysis of the data; in Section 3 we describe model

specification and we perform a model selection analysis through the classical AIC and BIC information criteria. In Section 4 we conduct an out-of-sample forecasting analysis and in Section 5 we provide some concluding remarks.

2 Preliminary analysis

Daily prices for Bitcoin, Ethereum, Litecoin and Monero are retrieved from the website <https://coinmarketcap.com/> relative to the period January 2016 - October 2019.

Cryptocurrencies are subject to changes in price due to imbalance between supply and demand but do not have any natural mechanism that produces dividends and returns. For this reason we assume in what follows that their price can be modeled as an arithmetic random walk. We perform the Augmented Dickey Fuller (ADF) and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests on prices and price differences for the four cryptocurrencies under analysis: the former tests the null hypothesis of unit root while the latter tests the null of trend stationarity against the alternative of unit root. Both tests agree in rejecting stationarity for price levels while they do not reject stationarity of price differences. Therefore, we can conclude that cryptocurrencies under analysis are integrated of order one ($I(1)$) while their differences are stationary ($I(0)$).

Table 1: Summary statistics of daily price differences

	Bitcoin	Ethereum	Litecoin	Monero
Min.	-2329.300	-238.230	-52.530	-94.530
Q_1	-1709.500	-129.654	-32.815	-54.526
Median	-1368.915	-103.628	-22.503	-45.910
Mean	5.619	0.130	0.037	0.037
Q_3	-1309.316	-77.175	-21.024	-41.694
Max.	3608.200	151.210	101.960	91.960
St. Dev.	324.435	21.495	6.445	9.425
Skewness	0.714	-0.670	3.662	-0.497
Kurtosis	25.609	27.452	69.088	29.307
ADF-Test p -value	0.001	0.001	0.001	0.001
KPSS-Test p -value	0.100	0.100	0.100	0.100

Summary statistics, ADF and the KPSS tests of daily price differences for Bitcoin, Ethereum, Litecoin and Monero from January 1, 2016 to October 15, 2019.

Table 1 summarizes the statistical parameters for the price differences of

analyzed series, and Figure 1 represents the behavior of the corresponding prices.

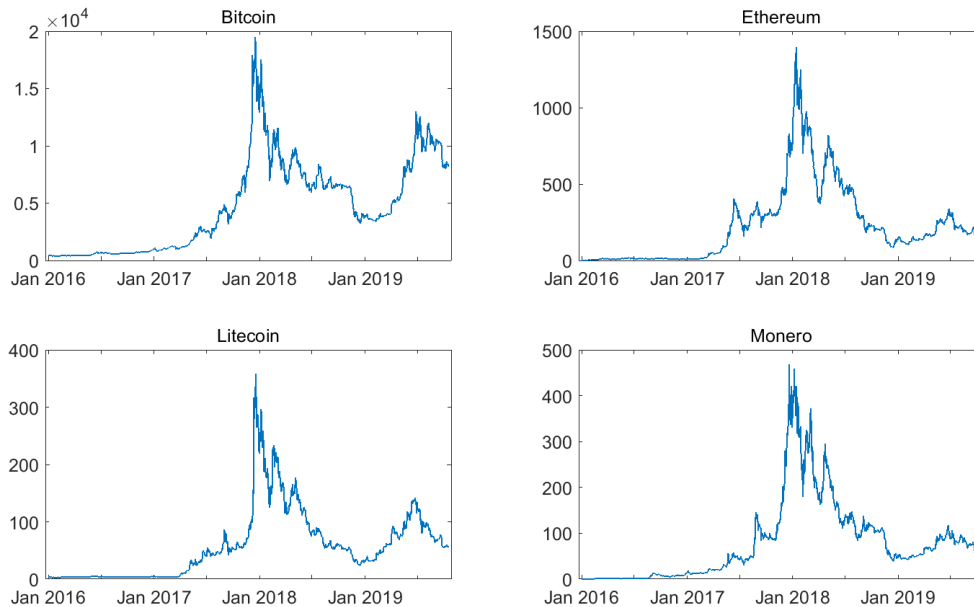


Figure 1: Price behaviour for Bitcoin (top-left), Ethereum (top-right), Litecoin (bottom-left) and Monero (bottom-right) from January 1, 2016 to October 15, 2019.

3 Analysis of regime switch

Regime switching models are an important class of models used in financial and non-financial applications as regimes identification carries relevant information for most applications. For example, in speech recognition the regimes, described by hidden states, represent the words reconstructed from vocal emissions. In finance, regime switches capture the intuition that markets and economies in general might change from different phases, for example so-called bear or bull markets.

Though regime changes have been previously proposed by Quandt (1958) and Goldfeld and Quandt (1973), it was James Hamilton who introduced the comprehensive theory of Hidden Markov Models and also suggested an estimation procedure based on Bayesian filter, see Hamilton (1989, 1990). HMMs describe the regimes through a hidden Markov chain and the corresponding transition matrix is estimated together with the state-dependent model parameters. In this paper we consider HMMs with fixed transition probabilities. Extensions to time varying transition probabilities have been

proposed in the literature, see Diebold et al. (1994); Durland and McCurdy (1994) and Focardi et al. (2019); however, available time series of cryptocurrencies are not sufficiently long to allow the estimation of HMMs in this generalized framework.

Let us assume that each cryptocurrency l , for $l \in \{1, 2, 3, 4\}$, is governed by a hidden Markov chain $Y_l = \{Y_{l,t}, t \in \mathbb{N}\}$ with k states $s_i = 1, 2, \dots, k$ and denote with $\mathbf{T}_l$ the corresponding transition matrix:

$$\mathbf{T}_l = \begin{bmatrix} p_{l,11} & \cdots & p_{l,1k} \\ \vdots & \ddots & \vdots \\ p_{l,k1} & \cdots & p_{l,kk} \end{bmatrix}, \quad \sum_{j=1}^k p_{l,ij} = 1,$$

where $p_{l,ij} := P(Y_{l,t} = s_j | Y_{l,t-1} = s_i)$ is the probability of transition from state s_i at time $t-1$ to state s_j at time t , for any $t \in \mathbb{N}$.

In this framework it is also possible to derive the average sojourn time $AT_{l,j} := 1/(1 - p_{l,jj})$ of stock l in state j , that is, the average time that cryptocurrency l spends in state j .

In the special case where $Y_l = Y$, for $l = 1, 2, 3, 4$ the analysed cryptocurrencies share common regimes and transition probabilities.

3.1 Univariate regime modeling

We consider two possible models for the first differences $d_{l,t} := P_{l,t} - P_{l,t-1}$ of each cryptocurrency: the generalized white noise (GWN) and the autoregressive AR(1) model. The former is described by the following equation:

$$d_{l,t} = \mu_{l,0}(s) + \sigma_l(s)\epsilon_{l,t}, \quad l = \{1, 2, 3, 4\}, \quad t \geq 0, \quad (1)$$

where both coefficients $\mu_{l,0}(s)$ and $\sigma_l(s)$ depend on the state s and $\epsilon_{l,t}$ is a Gaussian white noise with unit variance.

The AR(1) model is written as

$$d_{l,t} = \mu_{l,0}(s) + \mu_{l,1}(s)d_{l,t-1} + \sigma_l(s)\epsilon_{l,t}, \quad l = \{1, 2, 3, 4\}, \quad t \geq 0, \quad (2)$$

where $d_{l,t}$ and $d_{l,t-1}$ are the differences of prices at time t and $t-1$ respectively, the $\mu_{l,0}(s)$, $\mu_{l,1}(s)$ and $\sigma_l(s)$ are the model coefficients that depend on the state s and $\epsilon_{l,t}$ is a Gaussian white noise with unit variance.

Models are fitted to available data through the Maximum Likelihood Estimation method by applying the package *MS_Regress* for Matlab[®] (Perlin, 2015) that applies the recursive filter proposed in Hamilton (1990); the estimation is performed for up to 4 states. Let's observe that the above models are conditionally Gaussian but the nature of HMM modelling makes unconditional distribution of price differences not Gaussian.

In order to visualize the different regimes, for $l = 1, 2, 3, 4$ and $k = 2, 3, 4$, we claim that cryptocurrency l is in state j when the smoothed state probability $\hat{p}_{l,j}$, computed by applying Hamilton (1990) filter, is larger than 0.5. In each model and cryptocurrency, regimes are ranked according to increasing values of the volatility parameter estimates. Binary variables are suitably defined for each state which are plotted in Figures 2 and 3, respectively for the GWN-HMM and the AR-HMM specifications. Each row corresponds to a specific cryptocurrency while each column to the number of states $k = 2, 3, 4$. Many takeaways are possible by looking at the above pictures:

- it is evident that a fourth state is useless for most cases when $k = 4$: the dynamics appear to switch among only three states; few exceptions are Monero, according to the GWN, and Bitcoin according to the AR(1) specifications.
- the two model specifications mostly agree in the state identification except in the $k = 4$ case.
- there is a large agreement between the four cryptocurrencies states when $k = 2$ (low and high volatility regimes) for both models;
- the consistency of cryptocurrencies states deteriorates with increasing number of states: for $k = 3$ regime switches still show commonalities, but for Monero;
- when $k = 4$ the analysis of common regimes is confounding since when only three regimes are relevant and their relative ranking may be different; for example the third state of Bitcoin is surely common to the fourth state of Ethereum in the GWN case.

Comparisons are given in terms of Akaike (AIC) and Bayesian (BIC) information criteria in Table 2. The results clearly evidence the presence of multiple regimes for each cryptocurrency in both models; the major gain in terms of AIC-BIC values is obtained when a second regime is accounted for while lesser improvements are obtained by adding additional regimes. Optimal AIC and BIC values are obtained when 3-4 regimes are considered; corresponding values suggest that the introduction of an autoregressive factor does not improve data description, but for Ethereum.

3.2 Multivariate regime modeling

In order to confirm our conjecture that regimes are common across the analysed cryptocurrencies, as evidenced by Figure 2, correlations among cor-

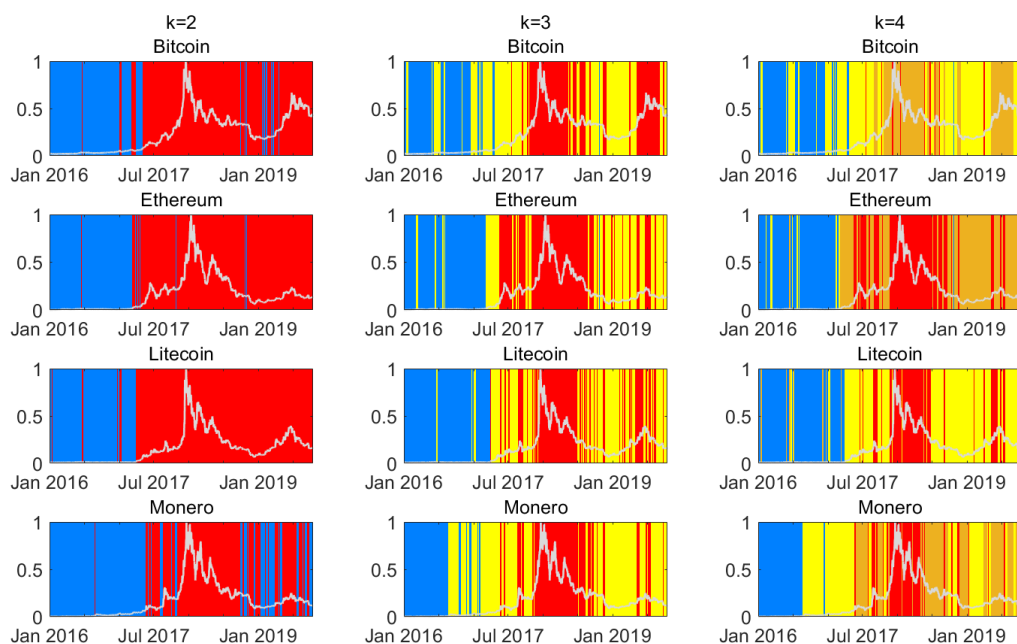


Figure 2: Cryptocurrency states for the GWN models with $k = 2, 3, 4$.

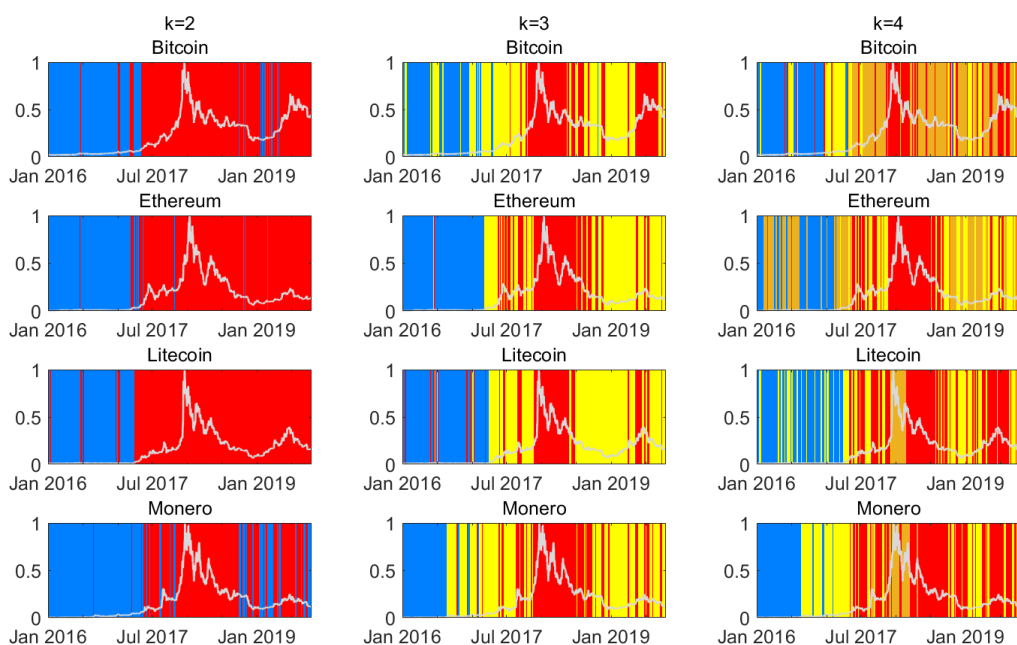


Figure 3: Cryptocurrency states for the AR(1) models with $k = 2, 3, 4$.

Table 2: AIC and BIC model selection analysis

k	Par.	Bitcoin		Ethereum		Litecoin		Monero	
		AIC	BIC	AIC	BIC	AIC	BIC	AIC	BIC
Panel a: GWN									
1	1	19919	19924	12411	12417	9080	9085	10131	10136
2	6	17411	17443	9585	9617	5772	5803	7435	7467
3	12	16741	16803	8944	9007	4850	4913	6304	6367
4	20	16748	16853	8776	8881	4697	4802	6278	6383
Panel b: AR(1)									
1	2	19906	19916	12403	12413	9071	9081	10105	10116
2	8	17412	17454	9556	9598	5867	5908	7404	7446
3	15	16764	16843	8799	8878	5187	5266	6431	6509
4	24	16700	16826	8656	8781	4702	4828	6291	6417

Akaike and Bayesian Information criteria for regime switching models with up to 4 states: generalized white noise (Panel a) and autoregressive AR(1) (Panel b). Optimal values of AIC and BIC are evidenced in bold. In column *Par.* the total number of estimated parameters is reported for each model specification.

responding regimes are summed up¹ in Tables 3 to 5 for $k = 2, 3, 4$ states. Correlation values evidence that regimes are highly correlated for up to three states: the highest values are obtained for the two regimes case while, when a third regime is allowed, the lowest and highest volatility regimes are more correlated than the medium volatility scenario. The introduction of a fourth state clearly undermines the regimes commonality.

Overall, the empirical results suggest that at most three common states may be considered for the basket of cryptocurrencies under investigation. A fourth state may be relevant as an added factor to the dynamics description of the individual cryptocurrencies rather than to the whole basket.

Table 3: Correlation matrix of the smoothed probabilities of GWN model with $k = 2$

		Bitcoin	Ethereum	Litecoin	Monero
S_1-S_2	Bitcoin	1.0000	0.8390	0.7733	0.7142
	Ethereum	0.8390	1.0000	0.8423	0.6911
	Litecoin	0.7733	0.8423	1.0000	0.5983
	Monero	0.7142	0.6911	0.5983	1.0000

¹Regime correlations are reported only for the GWN model; results are analogous in the AR(1) model specification and are available upon request.

Table 4: Correlation matrix of the smoothed probabilities of GWN model with $k = 3$

		Bitcoin	Ethereum	Litecoin	Monero
S_1	Bitcoin	1.0000	0.8303	0.8066	0.6954
	Ethereum	0.8303	1.0000	0.8889	0.7253
	Litecoin	0.8066	0.8889	1.0000	0.7589
	Monero	0.6954	0.7253	0.7589	1.0000
S_2	Bitcoin	1.0000	0.4818	0.5731	0.4883
	Ethereum	0.4818	1.0000	0.6537	0.6021
	Litecoin	0.5731	0.6537	1.0000	0.5579
	Monero	0.4883	0.6021	0.5579	1.0000
S_3	Bitcoin	1.0000	0.5847	0.6890	0.6723
	Ethereum	0.5847	1.0000	0.6823	0.7850
	Litecoin	0.6890	0.6823	1.0000	0.7077
	Monero	0.6723	0.7850	0.7077	1.0000

Table 5: Correlation matrix of the smoothed probabilities of GWN model with $k = 4$

		Bitcoin	Ethereum	Litecoin	Monero
S_1	Bitcoin	1.0000	0.8283	0.8118	0.5997
	Ethereum	0.8283	1.0000	0.8965	0.6468
	Litecoin	0.8118	0.8965	1.0000	0.6377
	Monero	0.5997	0.6468	0.6377	1.0000
S_2	Bitcoin	1.0000	-0.0725	0.5934	0.2165
	Ethereum	-0.0725	1.0000	-0.1597	-0.0408
	Litecoin	0.5934	-0.1597	1.0000	0.1596
	Monero	0.2165	-0.0408	0.1596	1.0000
S_3	Bitcoin	1.0000	0.5546	0.4720	0.3096
	Ethereum	0.5546	1.0000	0.4170	0.3374
	Litecoin	0.4720	0.4170	1.0000	0.2091
	Monero	0.3096	0.3374	0.2091	1.0000
S_4	Bitcoin	1.0000	-0.0373	0.1607	0.0871
	Ethereum	-0.0373	1.0000	-0.0814	-0.0769
	Litecoin	0.1607	-0.0814	1.0000	0.7217
	Monero	0.0871	-0.0769	0.7217	1.0000

If regimes were common to the whole basket of cryptocurrencies a multivariate extension of model specification in (1) and (2) would be a suitable description for the price change dynamic.

The Multivariate Generalized White Noise (MGWN) and Vector Autoregressive of order one (VAR(1)) specifications are given by Equations (3) and (4) respectively.

$$\mathbf{d}_t = \boldsymbol{\mu}_0(s) + \epsilon_t \quad (3)$$

$$\mathbf{d}_t = \boldsymbol{\mu}_0(s) + \mathbf{A}_1(s)\mathbf{d}_{t-1} + \epsilon_t \quad (4)$$

where $\mathbf{d}(t) = (d_1, d_2, d_3, d_4)$ is the vector of price differences, $\boldsymbol{\mu}_0 = (\mu_{1,0}, \mu_{2,0}, \mu_{3,0}, \mu_{4,0})$ is the vector of intercepts, $\mathbf{A}_1$ is the coefficients matrix and $\epsilon(t) = (\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4)$ is the error process with $\epsilon \sim \mathcal{N}(0, \Sigma(s))$. State binary variables are defined as in the univariate case and plotted in Figure 4 to visualize regime changes for our basket of cryptocurrencies. Again, the outcomes for the two model specifications are essentially indistinguishable.

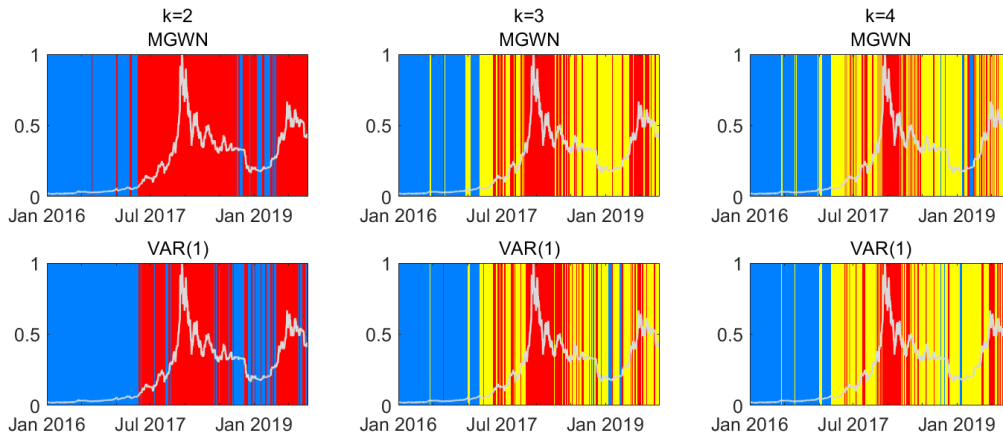


Figure 4: Cryptocurrencies states for MGWN and VAR(1) models, with $k = 2, 3, 4$.

In Table 6 the AIC and BIC values are summed up for up to $k = 4$ states; the optimal specification is obtained by assuming a multivariate GWN with three states. This outcomes finally confirm the presence of three common states across cryptocurrencies and the absence of a relevant autoregressive component in the price difference dynamics of our basket.

Table 6: AIC and BIC multivariate model selection analysis

Multivariate GWN				VAR(1)		
	Par.	AIC	BIC	Par.	AIC	BIC
$k = 1$	4	49191	49212	20	48963	49068
$k = 2$	30	42007	42205	62	41576	41986
$k = 3$	48	38959	39277	96	40258	40893
$k = 4$	68	39774	40225	132	39665	40539

Akaike and Bayesian Information criteria for regime switching models with up to 4 states: multivariate generalized white noise and vector autoregressive VAR(1). Optimal values of AIC and BIC are evidenced in bold. In columns *Par.* the total number of estimated parameters is reported for each model specification.

4 Out-of-sample analysis

In this section we analyze the out-of-sample performance of the univariate and multivariate GWN models² which have been described in (1) and (3). Precisely, the forecasts of the differences are obtained on moving windows of 1284 observations, for both models, and the root mean squared error RMSE between predicted and real values is adopted as a measure for the overall quality of the forecasts on the remaining 100 observations, considered as test dates. Table 7 displays the RMSE of forecasts by letting the number of regimes change from $k = 1$ to $k = 4$, respectively for the GWN (Panel a) and the multivariate GWN (Panel b) model specification. The introduction of possible regime switches clearly improves the overall quality of forecasts, as expected. The best forecasting performance, when cryptocurrencies are modeled individually, is obtained with a three regimes HMM for Bitcoin, Ethereum and Monero while two states best predict Litecoin price changes. As for the multivariate case, when common regimes are forced, the forecasting performance is optimized for $k = 2$ number of states. Similar outcomes are obtained when the accuracy of prediction is measured via the correlation coefficient between predicted and real values as suggested in West (2006). Overall, model selection through the optimization forecasting performance leads to a more conservative choice for the number of regimes to be taken into account with respect to the in-sample AIC and BIC information criteria. This results may be interpreted by arguing that the presence of three single or two common regimes is stable in time while additional regimes may be relevant only for specific sub-periods.

²The analysis has been carried out also for VAR models with analogous qualitative findings; details are not reported here since it has been shown above that the autoregressive component is not particularly relevant.

Table 7: RMSE between predicted and real values

	Bitcoin	Ethereum	Litecoin	Monero
Panel a: GWN				
$k = 1$	374.024	9.223	3.738	3.607
$k = 2$	374.371	9.163	3.697	3.624
$k = 3$	371.198	9.144	3.709	3.589
$k = 4$	374.069	9.168	3.724	3.598
Panel b: Multivariate GWN				
$k = 1$	374.024	9.223	3.738	3.607
$k = 2$	373.007	9.180	3.719	3.584
$k = 3$	372.764	9.200	3.725	3.600
$k = 4$	373.080	9.180	3.720	3.596

5 Concluding remarks

In this paper we prove that cryptocurrencies evidence a switching dynamics for both mean returns and volatility. Besides, we find a high correlation between probabilities of being in the same state across all the series and for both model specifications considered in this note. This finding suggests the presence of common factors driving cryptocurrencies price differences and corresponding regimes. Finally, we analyze the out-of-sample performance and show that HMMs do have forecasting power for the four cryptocurrencies, both in terms of RMSE and correlations between predicted and true values of prices. It is worth noticing that no modeling or forecasting improvement is achieved when applying neither AR nor VAR models: hence, the autoregressive component is not particularly relevant and there is no spillover prediction effect within different cryptocurrencies. Different regimes are characterized by pairs of volatility-expected returns; the estimation results for up to 4 states are collected in the Appendix. The interpretation of regimes depends on the selected model for each cryptocurrency and would need separate discussion. Nevertheless, it is possible to draw some conclusions from the overall picture: when two common regimes are considered, see Table 11, they may be safely interpreted as a low volatility and a high volatility state where high volatility is associated with negative mean differences, while low volatility is associated with positive mean differences. The probability of changing regime is lower than the persistence probability for each of the states and, notably, the high volatile-low mean regime has dominated the stable state in term of average sojourn time within the time period under analysis. It is worth noticing that

when the two regimes are analysed on individual cryptocurrencies, Bitcoin evidences positive mean differences in both low and high volatility regimes, suggesting the interpretation of the second state as a bubble regime with extremely high volatility and gains, see Table 8.

If we allow a third regime, the low-volatility state is trailed with two high-volatility periods respectively associated to (high) positive mean gains and negative mean gains as displayed in Tables 9 and 12. Though in-sample selection was optimal for $k = 4$ states for Ethereum and Litecoin the interpretation of regimes is not intuitive when taking into account a fourth regime.

Similarly, forcing four common regimes in the multivariate case completely messes up with interpretation as it is expected from both the AIC-BIC information criteria and the out-of-sample analysis performed in the paper.

Overall, we may conclude that cryptocurrencies are characterized by two or three states, of which at least two may be interpreted as common to the whole basket of considered cryptocurrencies and possibly generalized to the whole asset class.

We believe that it would be interesting to give an economic interpretation of the regimes in terms of changing trust on cryptocurrencies. Indeed, it is well known that cryptocurrencies were introduced as a means of exchange but have been proven to function mostly as speculative investments, the two usages interfering with each other. Using cryptocurrencies as means of exchange requires a high level of trust that the cryptocurrency will not change its value i.e low absolute mean and low volatility. On the other hand, speculative investment requires a positive trend of prices, as investors keep on buying under the assumption that prices will increase. Therefore, it seems reasonable to argue that the regimes that we have identified may represent basic situations of trust, which in itself might be motivated by exogenous factors. The investigation of this potential relationship is in our future research agenda.

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A Parameter estimates

We detail the analysis in case of a purely random behavior for the price differences in each regime as no significant improvement is obtained with autoregressive models.

Table 8: Parameter estimates of GWN with $k = 2$

	s	$\hat{\mu}_0$	$\hat{\sigma}$	AT	Trans. Matrix	
Bitcoin	1	3.78	18.97	15.67	0.9362	0.0475
	2	7.03	430.31	21.06	0.0638	0.9525
Ethereum	1	0.03	0.75	17.92	0.9442	0.0349
	2	-0.19	27.56	28.69	0.0558	0.9651
Litecoin	1	0.00	0.09	49.16	0.9797	0.0088
	2	-0.11	7.85	113.59	0.0203	0.9912
Monero	1	0.00	1.03	18.62	0.9463	0.0810
	2	-0.07	14.87	12.34	0.0537	0.9190

Table 9: Parameter estimates of GWN with $k = 3$

	s	$\hat{\mu}_0$	$\hat{\sigma}$	AT	Trans. Matrix		
Bitcoin	1	2.04	8.97	13.57	0.9263	0.0526	0.0000
	2	1.56	115.85	11.02	0.0728	0.9093	0.0586
	3	-3.17	630.63	17.07	0.0009	0.0381	0.9414
Ethereum	1	0.05	0.51	15.86	0.9369	0.0443	0.0953
	2	-0.21	5.85	6.91	0.0264	0.8553	0.1449
	3	0.15	31.26	4.16	0.0367	0.1004	0.7598
Litecoin	1	0.00	0.09	129.56	0.9992	0.0312	0.0000
	2	-0.05	2.03	4.75	0.0001	0.7895	0.4248
	3	-0.01	11.79	2.35	0.0007	0.1793	0.5752
Monero	1	0.00	0.12	21.28	0.9530	0.0185	0.0000
	2	0.28	2.40	19.07	0.0470	0.9475	0.0729
	3	-0.37	18.34	13.72	0.0000	0.0340	0.9271

Table 10: Parameter estimates of GWN with $k = 4$

	s	$\hat{\mu}_0$	$\hat{\sigma}$	AT	Trans. Matrix			
Bitcoin	1	1.72	8.86	12.90	0.9225	0.0546	0.0090	0.0625
	2	7.98	111.85	11.69	0.0771	0.9145	0.0031	0.8372
	3	14.60	559.89	17.43	0.0001	0.0266	0.9426	0.0764
	4	-17.27	994.65	1.02	0.0003	0.0043	0.0453	0.0239
Ethereum	1	-0.02	0.49	21.94	0.9544	0.0032	0.0012	0.7203
	2	-0.04	1.57	1.00	0.0409	0.0000	0.0006	0.0000
	3	-0.20	7.21	26.84	0.0030	0.9627	0.0907	0.0002
	4	0.24	39.16	10.81	0.0016	0.0340	0.9075	0.2795
Litecoin	1	0.00	0.07	14.49	0.9310	0.0059	0.1387	0.0623
	2	-0.03	2.33	25.73	0.0123	0.9611	0.0171	0.1500
	3	0.06	4.49	1.49	0.0229	0.0078	0.3276	0.1866
	4	-0.05	17.29	2.51	0.0338	0.0251	0.5167	0.6012
Monero	1	0.03	0.09	48.08	0.9792	0.0237	0.1269	0.1423
	2	0.03	1.27	18.82	0.0012	0.9469	0.0000	0.2063
	3	0.02	5.08	5.34	0.0135	0.0159	0.8128	0.0945
	4	0.03	25.16	2.26	0.0060	0.0136	0.0602	0.5569

Table 11: Parameter estimates of MGWN with $k = 2$

	s	$\hat{\mu}_0$	$\hat{\sigma}$	AT	Trans. Matrix	
Bitcoin	1	9.57	220.91	5.53	0.8192	0.0623
	2	-3.89	363.15	16.06	0.1808	0.9377
Ethereum	1	0.11	8.70	5.53	0.8192	0.0623
	2	-0.26	30.68	16.06	0.1808	0.9377
Litecoin	1	0.00	7.01	5.53	0.8192	0.0623
	2	-0.02	7.35	16.06	0.1808	0.9377
Monero	1	0.03	1.96	5.53	0.8192	0.0623
	2	-0.03	11.41	16.06	0.1808	0.9377

Table 12: Parameter estimates of MGWN with $k = 3$

	s	$\hat{\mu}_0$	$\hat{\sigma}$	AT	Trans. Matrix		
Bitcoin	1	5.09	235.63	8.99	0.8887	0.0484	0.2666
	2	-6.08	269.59	6.95	0.0073	0.8561	0.1743
	3	6.16	485.61	2.27	0.1040	0.0955	0.5591
Ethereum	1	0.09	17.94	8.99	0.8887	0.0484	0.2666
	2	0.12	22.97	6.95	0.0073	0.8561	0.1743
	3	0.15	31.16	2.27	0.1040	0.0955	0.5591
Litecoin	1	0.04	4.58	8.99	0.8887	0.0484	0.2666
	2	-0.03	5.37	6.95	0.0073	0.8561	0.1743
	3	0.05	9.51	2.27	0.1040	0.0955	0.5591
Monero	1	0.06	6.75	8.99	0.8887	0.0484	0.2666
	2	-0.06	8.33	6.95	0.0073	0.8561	0.1743
	3	0.02	14.23	2.27	0.1040	0.0955	0.5591

Table 13: Parameter estimates of MGWN with $k = 4$

	s	$\hat{\mu}_0$	$\hat{\sigma}$	AT	Trans. Matrix			
Bitcoin	1	5.16	223.40	5.94	0.8318	0.2284	0.1717	0.2262
	2	-6.86	382.94	2.94	0.0292	0.6601	0.1191	0.1126
	3	5.48	477.68	2.67	0.0135	0.0179	0.6248	0.1148
	4	-5.05	545.41	2.20	0.1255	0.0936	0.0844	0.5464
Ethereum	1	0.17	6.54	5.94	0.8318	0.2284	0.1717	0.2262
	2	-0.12	19.57	2.94	0.0292	0.6601	0.1191	0.1126
	3	0.14	27.48	2.67	0.0135	0.0179	0.6248	0.1148
	4	-0.15	33.86	2.20	0.1255	0.0936	0.0844	0.5464
Litecoin	1	0.04	6.00	5.94	0.8318	0.2284	0.1717	0.2262
	2	-0.03	4.81	2.94	0.0292	0.6601	0.1191	0.1126
	3	0.04	7.98	2.67	0.0135	0.0179	0.6248	0.1148
	4	-0.04	13.91	2.20	0.1255	0.0936	0.0844	0.5464
Monero	1	0.04	4.44	5.94	0.8318	0.2284	0.1717	0.2262
	2	-0.04	9.25	2.94	0.0292	0.6601	0.1191	0.1126
	3	0.04	15.44	2.67	0.0135	0.0179	0.6248	0.1148
	4	-0.04	15.34	2.20	0.1255	0.0936	0.0844	0.5464